QC Story Project

Project Title:- To Eliminate the concern of door key sticky in MARAZZO vehicle.

Automotive Division, Nasik Plant , Scorpio / Xylo TCF Dept.

Current Status: Closed

	Plan	Actual
Start Date	10.11.2019	10.11.2019
Completion Date	31.12.2019	31.12.2019

Team	Name	Dept
Leader	**PAGAR.SATISH** Email : PAGAR.SATISH@mahindra.com	Scorpio / Xylo TCF
Members	Jitendra Baviskar	PVT
	Nilesh Kalekar	Scorpio / Xylo TCF
	Rajendra Suryawanshi	SQA
	Arvind Mistry	R&D
Dept Head	**Rahul Narkar**	Scorpio / Xylo TCF
Mentor	Shrikant Mulay	

1. Select the theme

1.1 Validate Selection
1.2 Define The problem
1.3 Form a team
1.4 Make A project Plan

1.1 Validate Selection Risk Assessment

No.	Success predictors	Yes (2)	Partial (1)	No (0)
1	**Project is a strategic priority, aligned organization goal (Big Y)**	2		
2	**Identify project area of focus :** Be sure there is no overlap with other MYB/MGB/MBB projects	2		
3	**Scope of the project is under the area of my influence**	2		
4	**Project scope is limited to only one product / one type of defect**	2		
5	**It is related to repetitive problem (chronic, not sporadic)**	2		
6	**Root cause is unknown and solution is unknown** (Type 4 problem)	2		
6	**Current level of the project is measured**	2		
7	**Potential Benefits of the project is identified**	2		
8	**Right vertical & horizontal team members are available**	2		
9	**Team members include technical experts**	2		
10	**Management is willing to commit serious resources to solutions**	2		
11	**Project is capable of completion within 4~6 months**	2		
12	**Project registration form is approved by HOD**	2		
	*** Any factor less than "2" creates a risk**			

Total:24

Success probability : High=22~24 Medium=19~21 Low=1~18

1.2 Define the problem

1. Brief Description of the problem:

To Eliminate the concern of key sticky MARAZZO vehicle.

Part contains :-

1) Key Fob
2) Lagger booger
3) Door lock

1.2 Define the problem

2. 4W 1H of the problem:

5W 1H	Details
What	**To Eliminate the concern of door key sticky from MARAZZO.**
When	In all models of MARAZZO since SOP.
Where	At CAI & LSP.
Who	Reported by the CAI inspectors & RFI Inspectors.
How Big	Daily 05 vehicles reported at CAI & RFI.

1.2 Define the problem

3. Problem details

S.NO	Check points	Details
1	Vehicle Model Selected for project	MARAZZO
2	Assembly / Part selected for project	Part Selected for Project :- Part contains :- 1) Key fob 2) Lagger booger 3) Door lock
3	Other Similar Models having the problem	No
4	Other similar models not having the problem:	SCORPIO, XYLO,U215,SC/DC,BMT _ No problem
5	Stages where the problem can be detected	CAI rattle stage
6	Processes under consideration (boundary/scope of project)	Supplier part 1) Key fob,2)lagger booger,3)Door lock Door lock & lagger booger fitment - Process.
7	Response type (Y):Variable/Attribute.	Attribute
8	Specification of the response:	Attribute (Sticky / No sticky)
9	Least count of Measurement System (if applicable)	No
10	Whether MSA is needed? Yes/No	NO

1.3 Form Team (involvement Matrix)

Instructions: Use the matrix to help you identify the people who should be involved in identifying cause, developing and implementing solutions.

Task	Details			
	Responsible for	Involved in	Consulted with	Informed About
Identifying causes	CFT	R&D & SQA	Supplier	Supplier
Validating the causes	MQA			
Identifying Solutions	MQA & R&D			
Selecting Solutions	MQA , R&D & Supplier	MQA, R&D & SQA		Supplier
Planning Implementation	R&D	R&D	Supplier	
Handling potential problems	MQA & R&D	R&D	Supplier	
Implementing solutions	R&D			
Monitoring results	MQA			CQA

Do we need involve anymore person in team? YES CQA (ANOVAC & LSP Auditors)

1.4 Project Plan

Project Planning

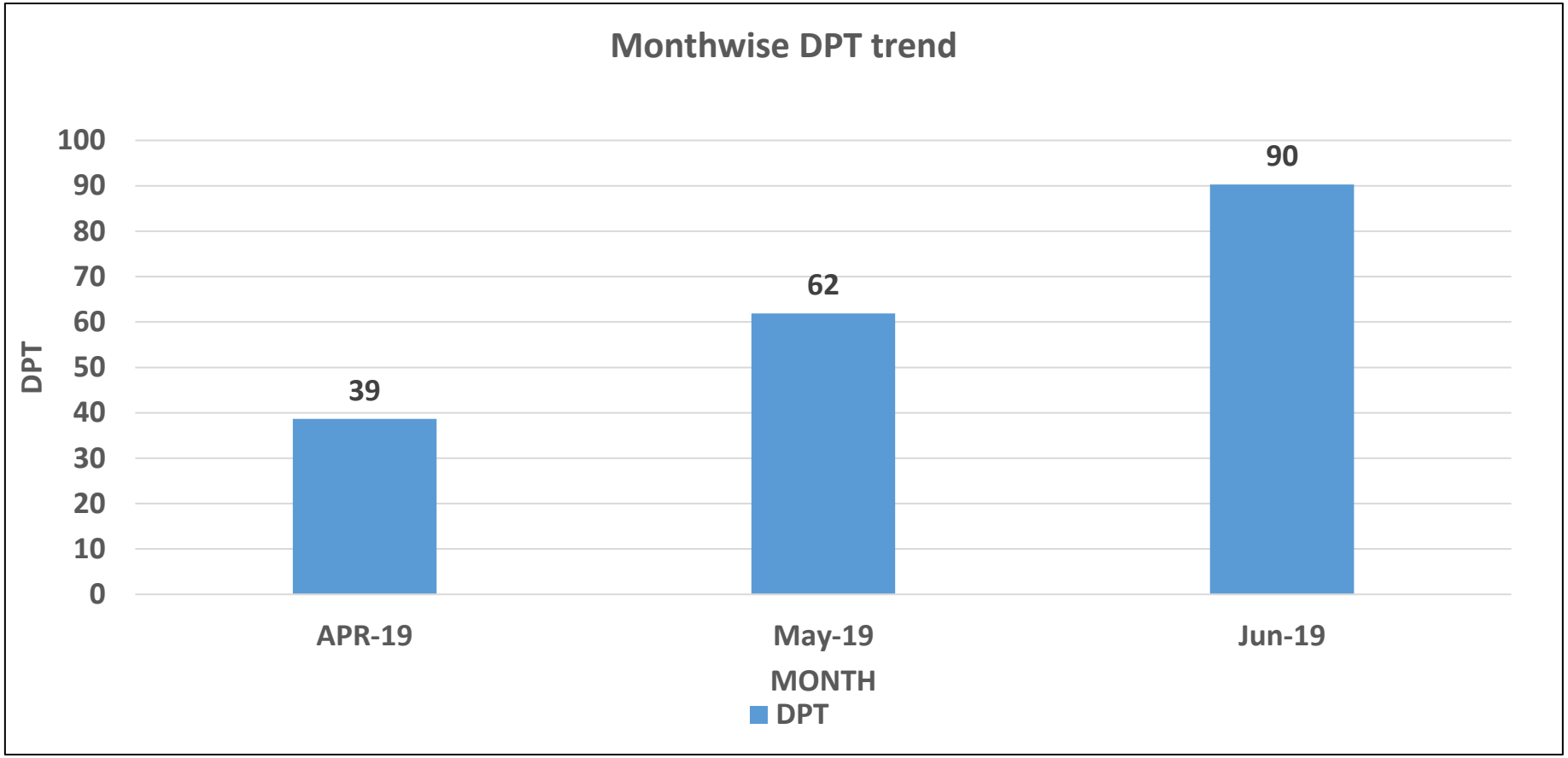
QC story Steps	Jun-19				July-19				Aug-19				Sep-19				OCt-19				Nov-19				Dec-19			
	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
1.Select the Theme																												
2.Justify the choice & Containment action																												
3.Understand the current situation																												
4.Select the target																												
5.Analyze																												
6.Implement the corrective measures																												
7.Verify the results and Effects																												
8.Standardize																												
9.Summarize and Future actions																												

Plan Actual

2 Justify the choice and containment action

2.1 Past trend of the problem
2.2 Pareto Diagram
2.3 Impact of the problem
2.4 Cost of Poor Quality
2.5 Containment action

2.1 Past Trend of the problem

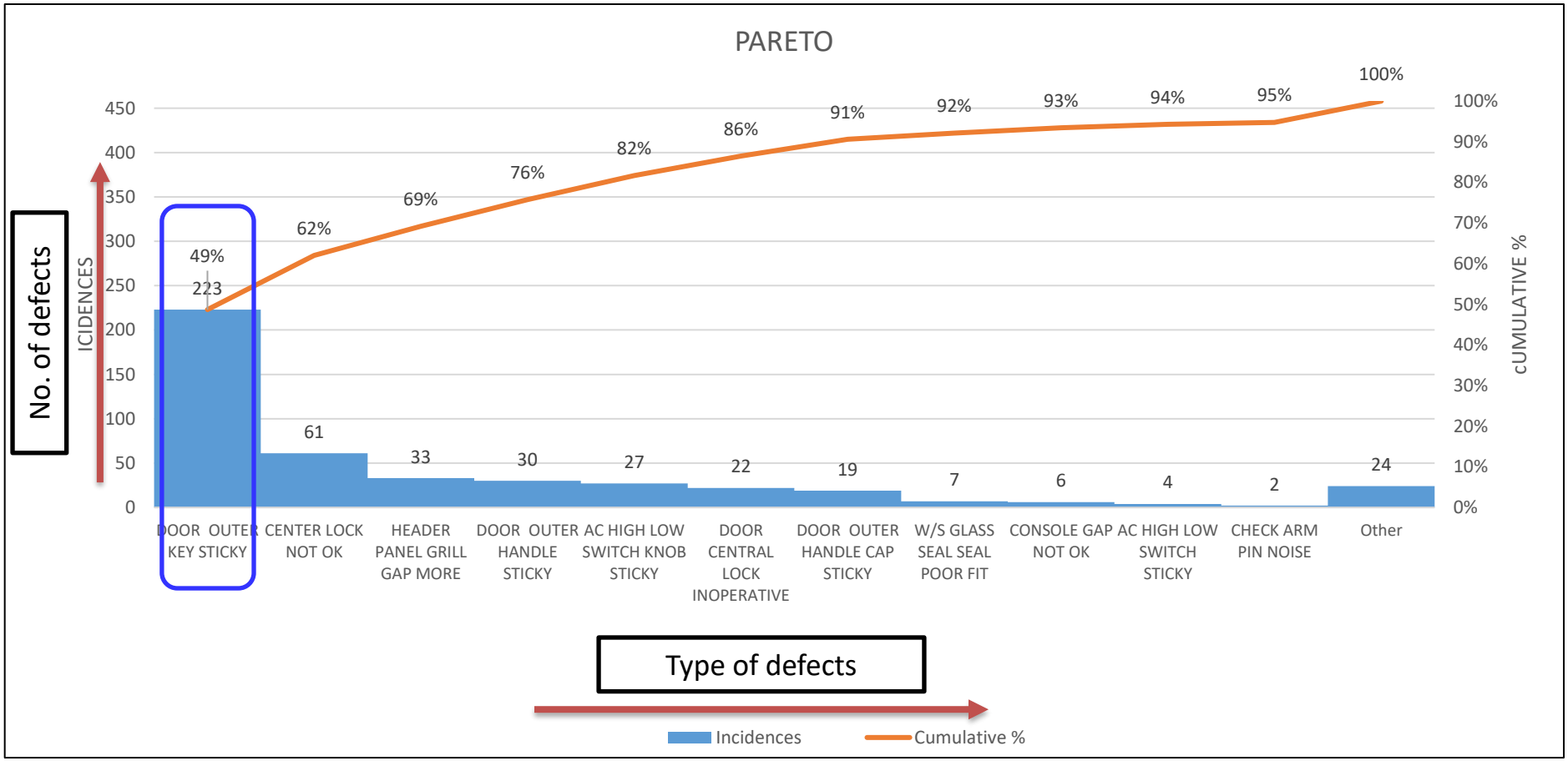


Inference:-

Average Monthly DPT for this concern – 64

2.2 Pareto Diagram

Top defect in CAI (Data Period : APR -19 to JUN-19)

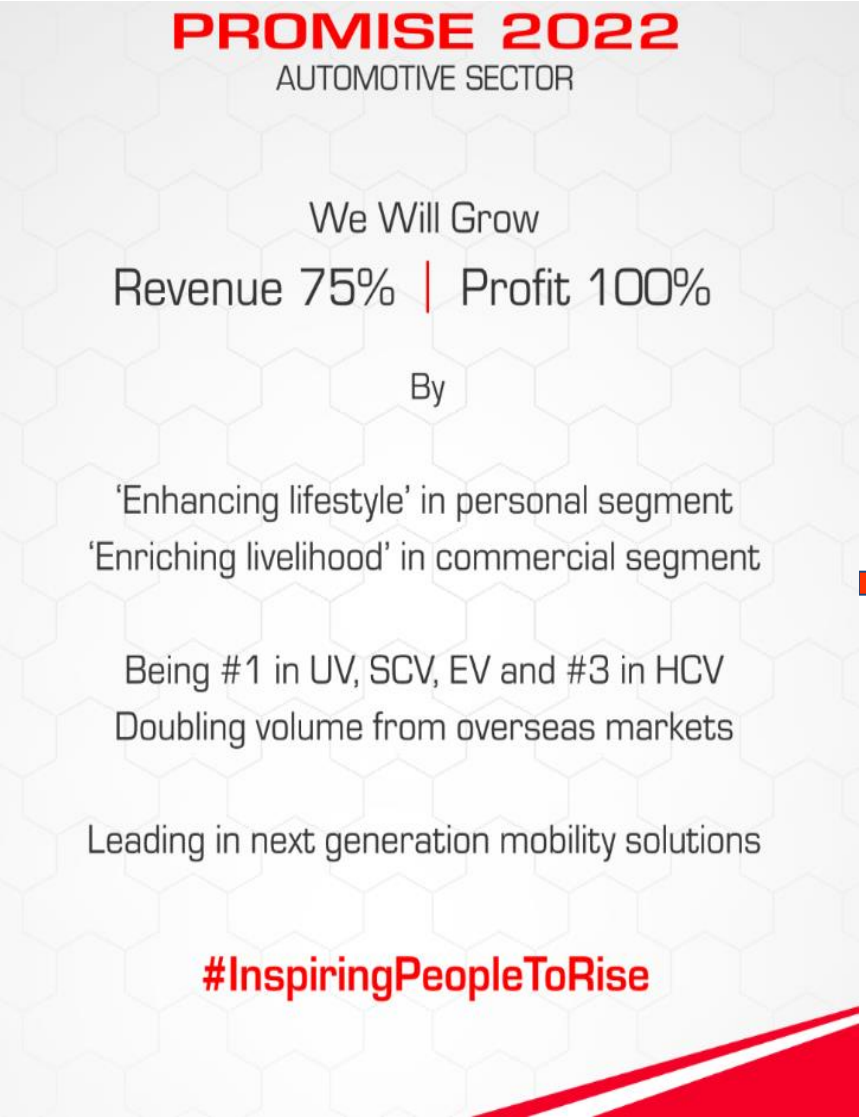


Inference:- Door key sticky is top contributing concern in at CAI.

Linkage with sector level goal

1. Tree Diagram

2. Justify the choice (reasons)



Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2.3 Impact of the problem

Impact on customer

1. Customer Dissatisfaction due to key sticky.

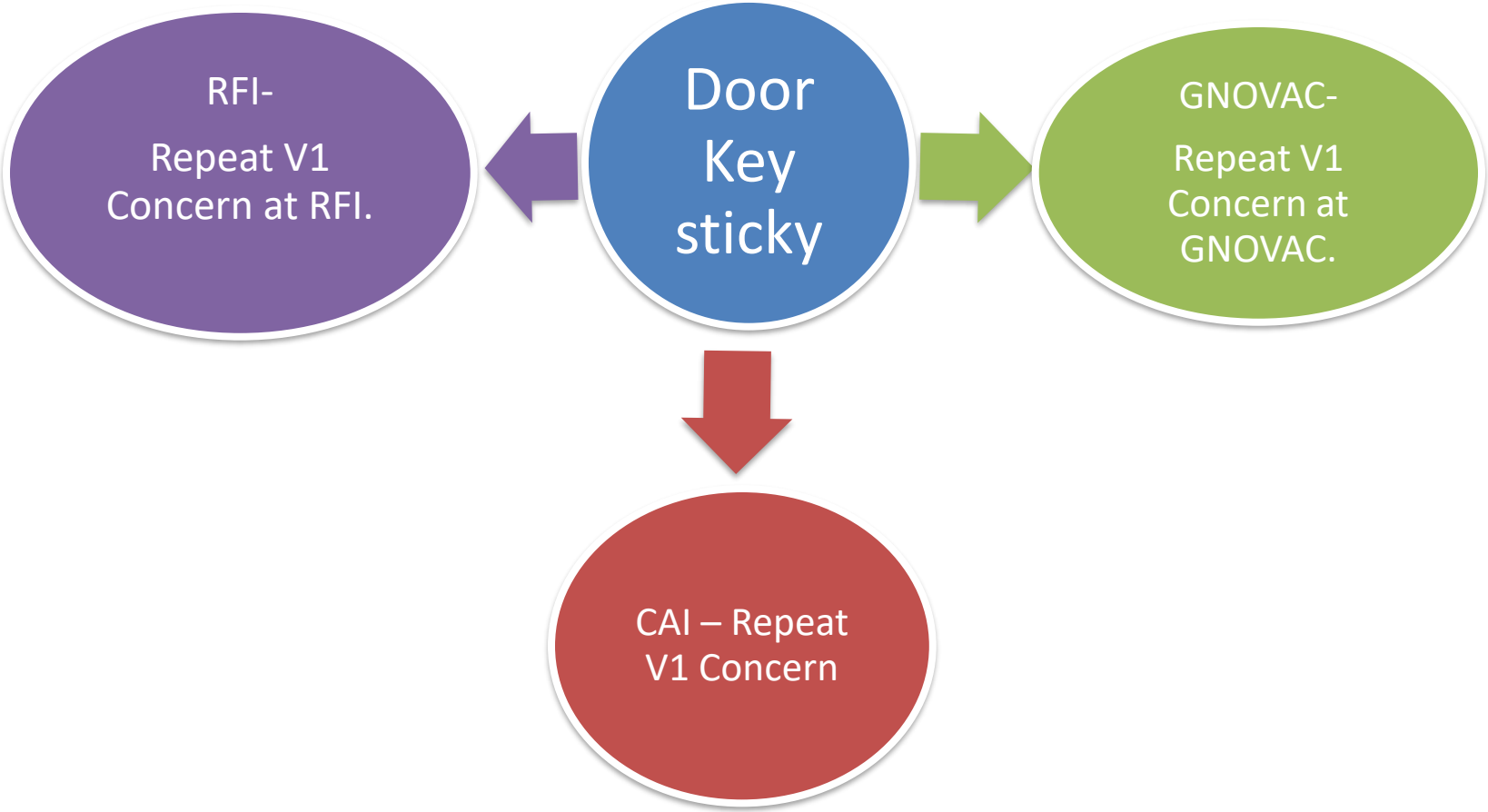
Impact on Mahindra brand

1. Customer Delightful experience with Mahindra Brand affected
2. Promise 2019 affected.

Impact on Department goal

1. Quality indicators affected
2. Rework cost increased
3. Rework time affected
4. Delivery of the vehicle affected

2.3 Impact of the problem



2.4 Cost of Poor Quality

Rework Cost estimation for door key sticky noise		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	10
Total Required Rework time per day	Min	100
Effective working time	Min	465
Operator required /day	nos	0.22
Approx Operator cost/ day	Rs	1360
Operator cost /concern	Rs	292.47
Avg no. of working days	days	25
Operator cost /month	Rs	7311.8
Operator Cost/year	**Rs**	**87741.9**
Total Rework cost	Rs / Year	87741.9

Productivity / Any other expected benefits from the project: :- 0.22 Person will save.

2.4 Cost of Poor Quality

Summary of total benefits expected from the project:

1. Improvement in Quality indicators
   (DPT Reduction, STR improvement, ANOVAC Score reduction, LSP DPT Score reduction)

2. Cost saving of 87741.09/- per year – as per COPQ

3. Rework time saving of 10 min per vehicle.

4. Quality Crusade targets improvement by First time right ,Every time right & Customer delight.

5. Helping to achieve First time Right launch of new product

2.5 Containment Action

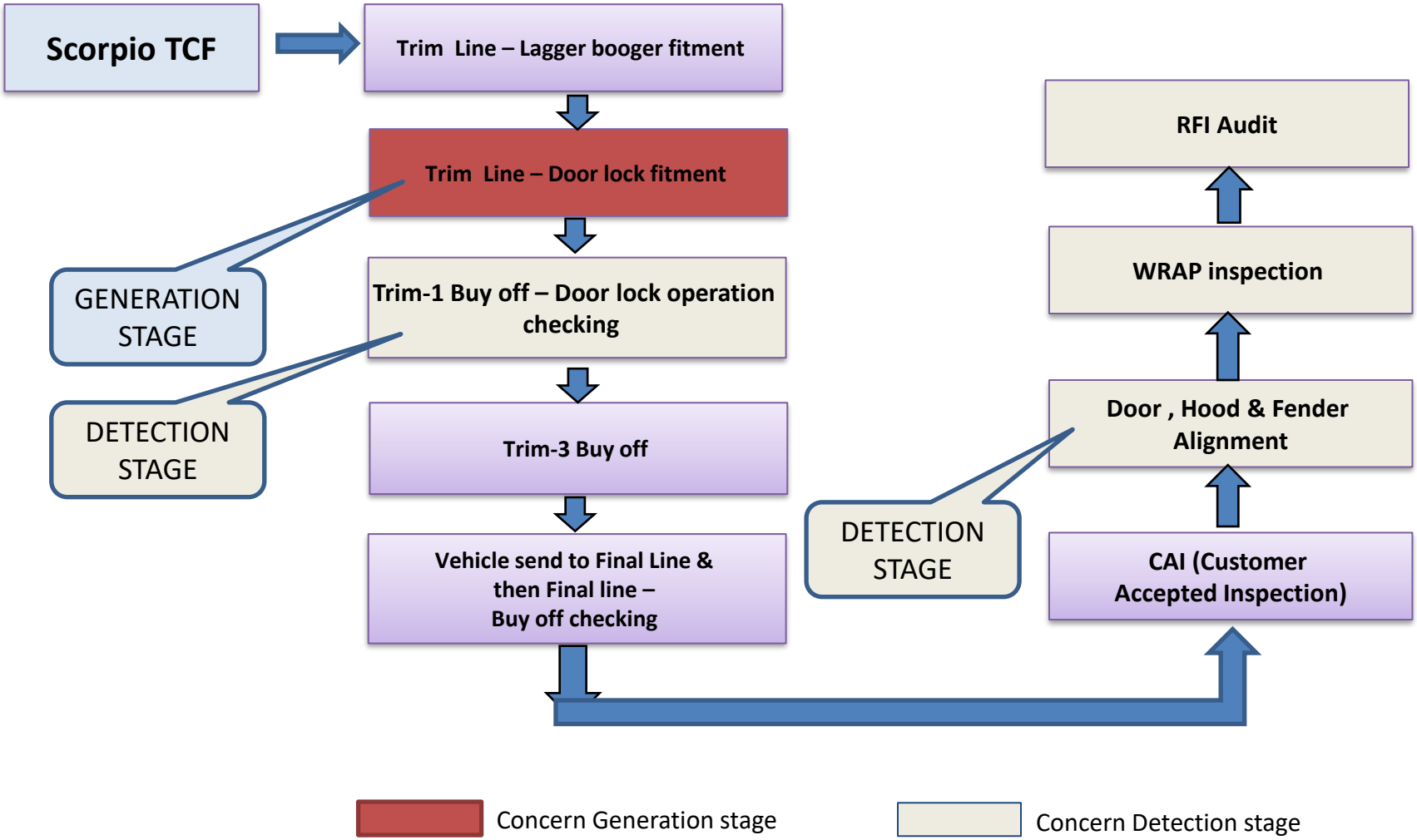
- Awareness given for defect identification & rework at CAI.

3. Understand the current Situation

3.1 Process Flow Diagram
3.2 Past data Analysis from different view point
3.3 Gemba Observations
3.4 Quick win Identification
3.5 Results of Quick wins

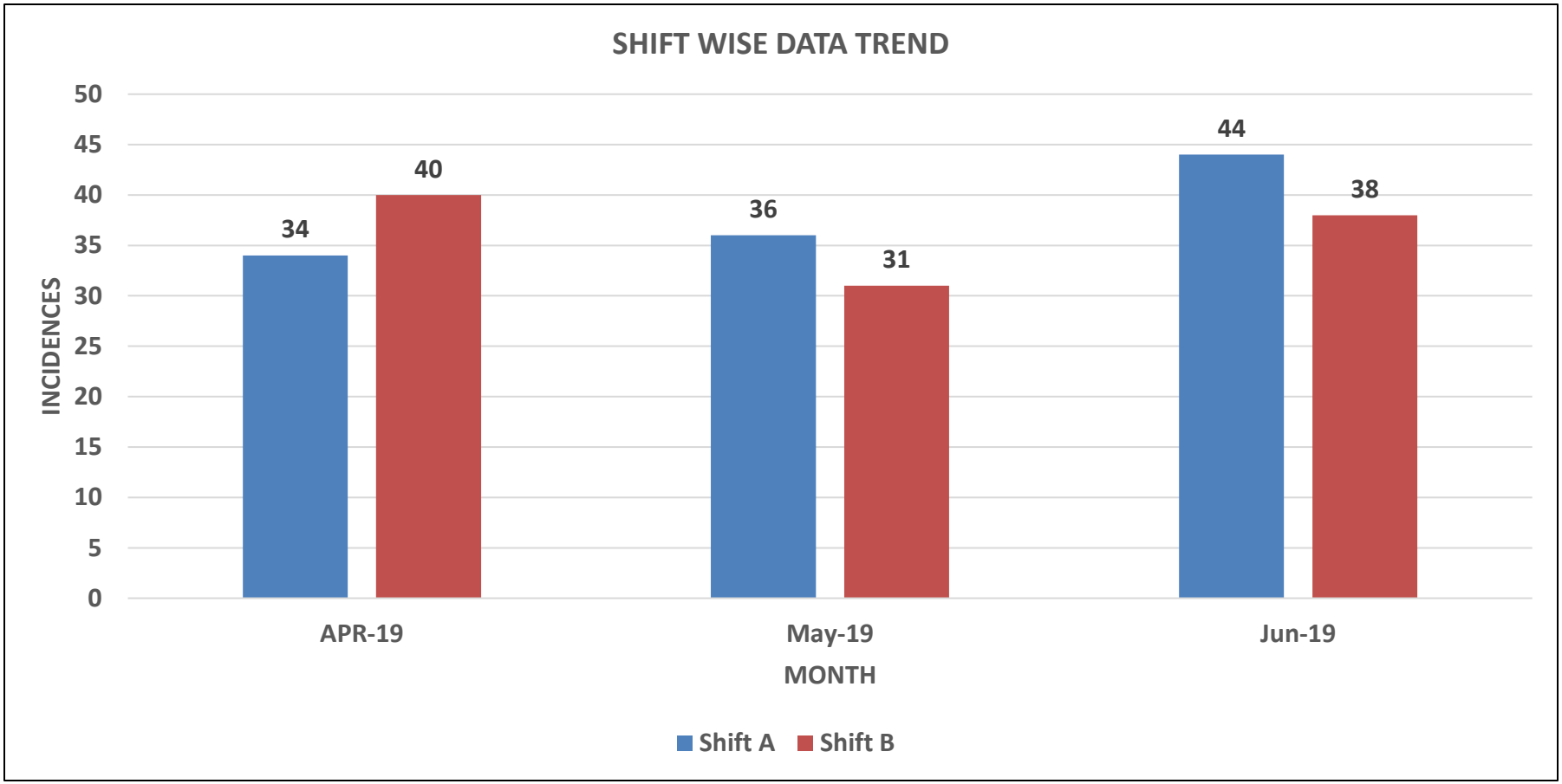
3.1 Process Mapping

Process flow for **door key sticky** concern in MARAZZO



3.2 Different View Point

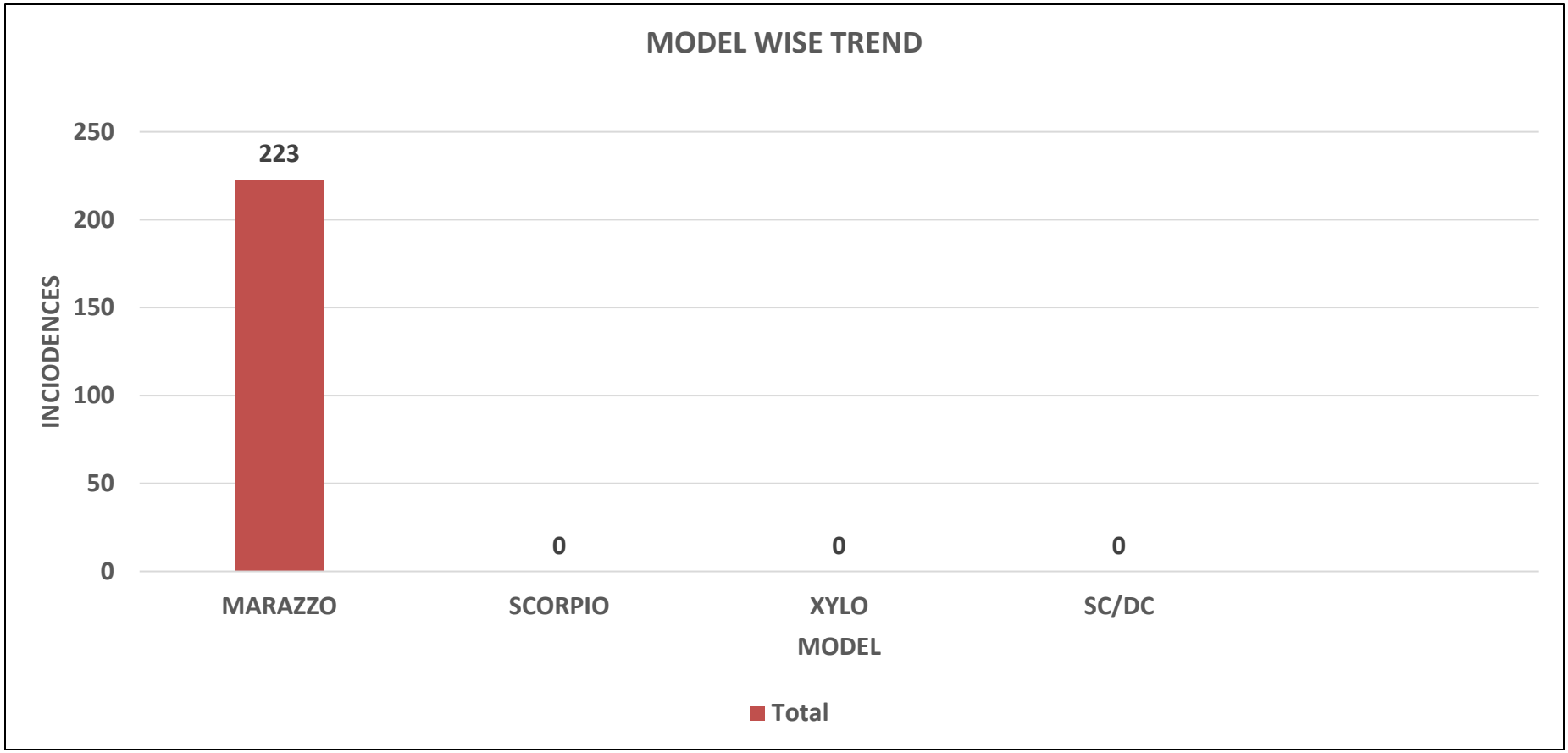
A. Graph of Time to Time Variation:



Inference :- No significant relation established.

3.2 Different View Point

B. Stream to Stream



Inference :- **Issue Only observed in MARAZZO vehicle.**

3.3 Gemba Observations

Gemba Observations By CFT :-

1) Associate not following the SOS of Door lock & lagger booger fitment.

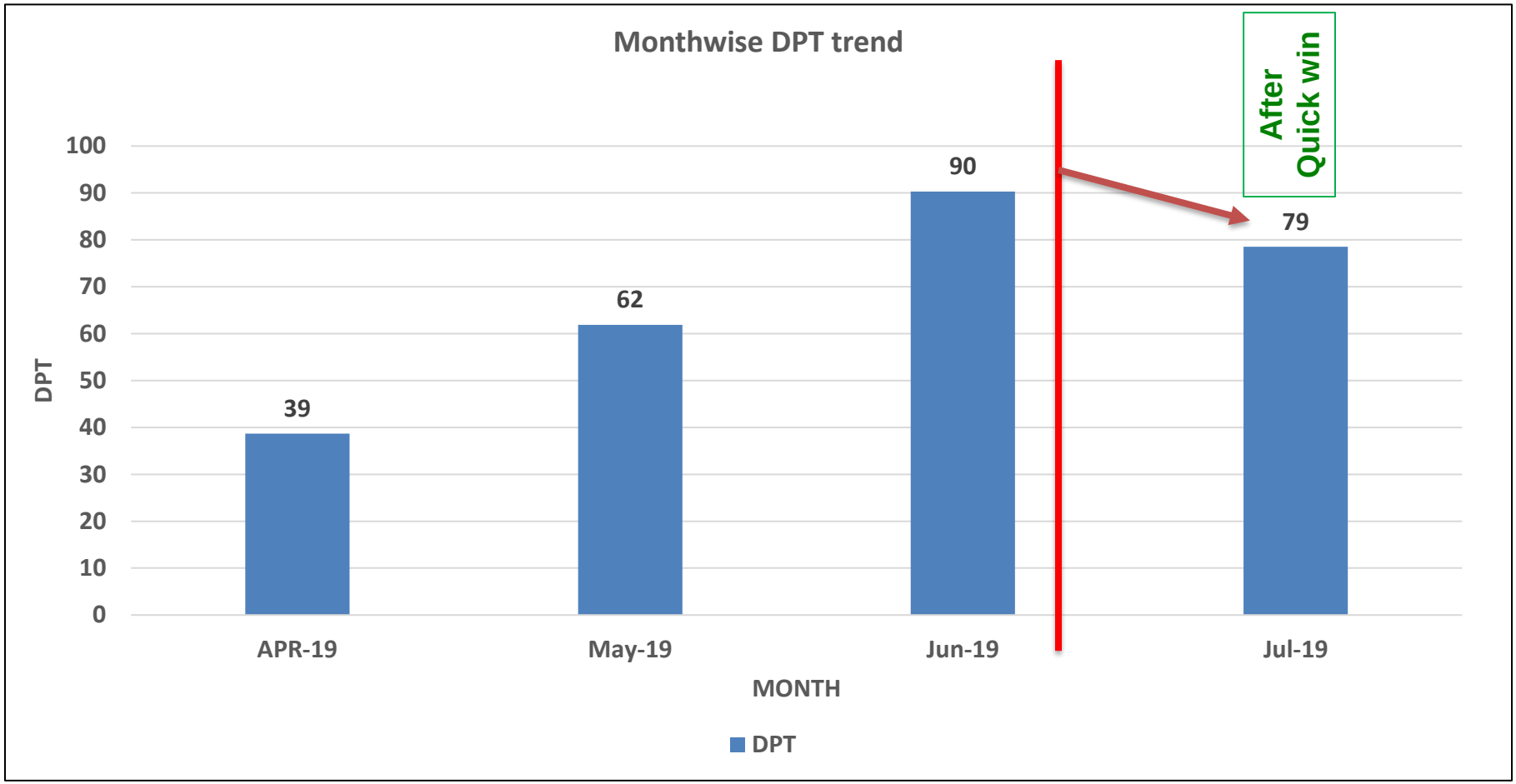
_ Door lock should be insert in outer handle cap from front side only as per SOS.

_ Key point added in SUCL.

3.4 Quick win Identification

Observation	Idea	Source	Easy to implement ?	Fast to implement?	Cheap to implement?	Easily Reversible?	Within Authority?	Implement Quick Win
Associate not following the SOS of lagger booger fitment	Awareness given to stage associate & point added in daily process audit.	**Gemba**	**YES**	**YES**	**YES**	**YES**	**YES**	**YES**

3.5 Results of Quick wins



Inference :

No significant improvement in results observed after quick win actions implemented.

3.5 Results of Quick wins

Hypothesis testing for confirmation of results after Quick win actions taken :-

Practical statement –

Defect rate of **door key sticky** (Y) was reduced after taking corrective measures through quick win.

Statistical Problem –
Ho: Defect rate after quick win >= Defect rate before quick win
Ha: Defect rate after Quick Win < Defect rate before quick win

Analysis: 2P Test -

3.5 Results of Quick wins

2-Sample % Defective Test for Group 1 vs Group 2
Summary Report

% Defective Test
Is Group 1 greater than Group 2?

The % defective of Group 1 is not significantly greater than the % defective of Group 2 (p > 0.05).

Individual Samples

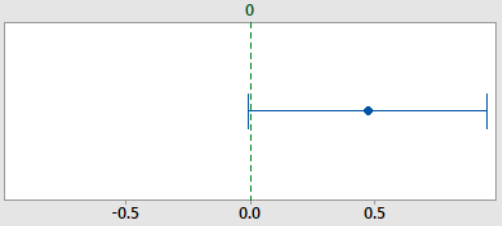
Statistics	Group 1	Group 2
Total number tested	8348	3102
Number of defectives	190	56
% Defective	2.28	1.81
90% CI	(2.01, 2.56)	(1.43, 2.25)

Difference Between Samples

Statistics	*Difference
Difference	0.47
90% CI	(-0.01, 0.95)

*Difference = Group 1 - Group 2

90% CI for the Difference
Is the entire interval above zero?

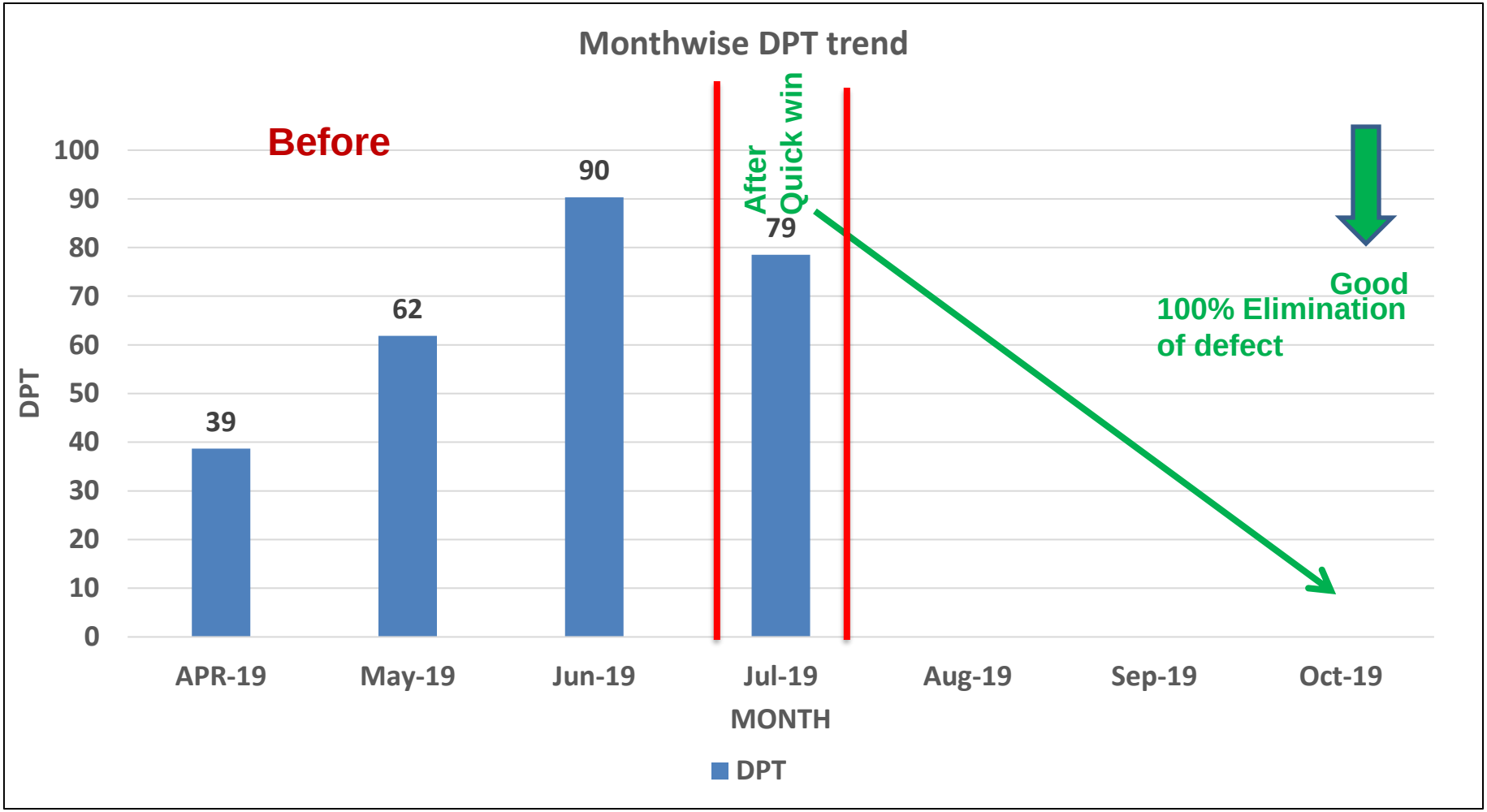


Comments

- Test: There is not enough evidence to conclude that the % defective of Group 1 is greater than Group 2 at the 0.05 level of significance.
- CI: Quantifies the uncertainty associated with estimating the difference from sample data. You can be 90% confident that the true difference is between -0.01% and 0.95%, and 95% confident that it is greater than -0.01%.

Statistical Conclusion – Accept the null hypothesis as P Value observed is > 0.05
Practical Conclusion – Defect rate of **door key sticky**(Y) was not reduced after taking corrective measures through quick win.

4. Select the target



5. Analyze

5.1 List the SSVs

5.2 Identify the tool

5.3 Collect the data

5.4 Test the Hypothesis

5.5 Conclusions of Analysis

- *Decide funneling needed or not*

5.6 Funneling to Root Cause

5.1 List down SSV's or Xs

A)

$$Y = f(x)$$

Response **SSV**

Y = Door key sticky (i.e BOB and WOW respectively)

X = X1. Assembly process **X2. Components in the assembly**

- **Initially CFT decided to find the root cause from Assembly process (X1)**

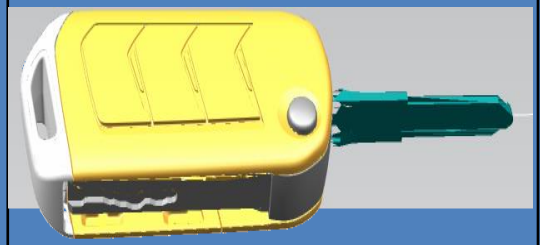
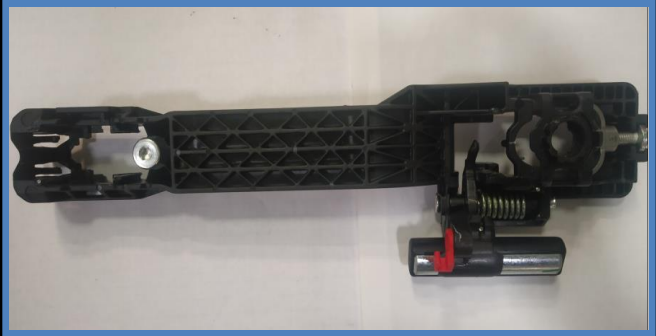
SIX SIGMA Tool identified

"COMPONENT SEARCH" or "MODIFIED COMPONENT SEARCH"

Based on assembly function

5.1 List down SSV's or Xs

- Team listed the parts in descending order, in which it will affect the response

Sr. No	Part Description		
1	Key fob	Lagger booger	Door lock
IMAGES			

5.2 Identify the tool

- Team listed the parts in descending order, in which it will affect the response

Sr. No	Part Description	Usage of parts after disassembly
1	Key Fob	**Can be reused**
2	Lagger booger	**Can be reused**
3	Door lock	**Can be reused**

Hence DOE Tool

"COMPONENT SEARCH"

was selected.

5.2 CRITERIA FOR SCALING – Y1 & GRAPH

- Team listed the parts in descending order, in which it will affect the response

Response (Y) – Door key sticky
SSV (X) - All 3 Parts combinations & assembly process.

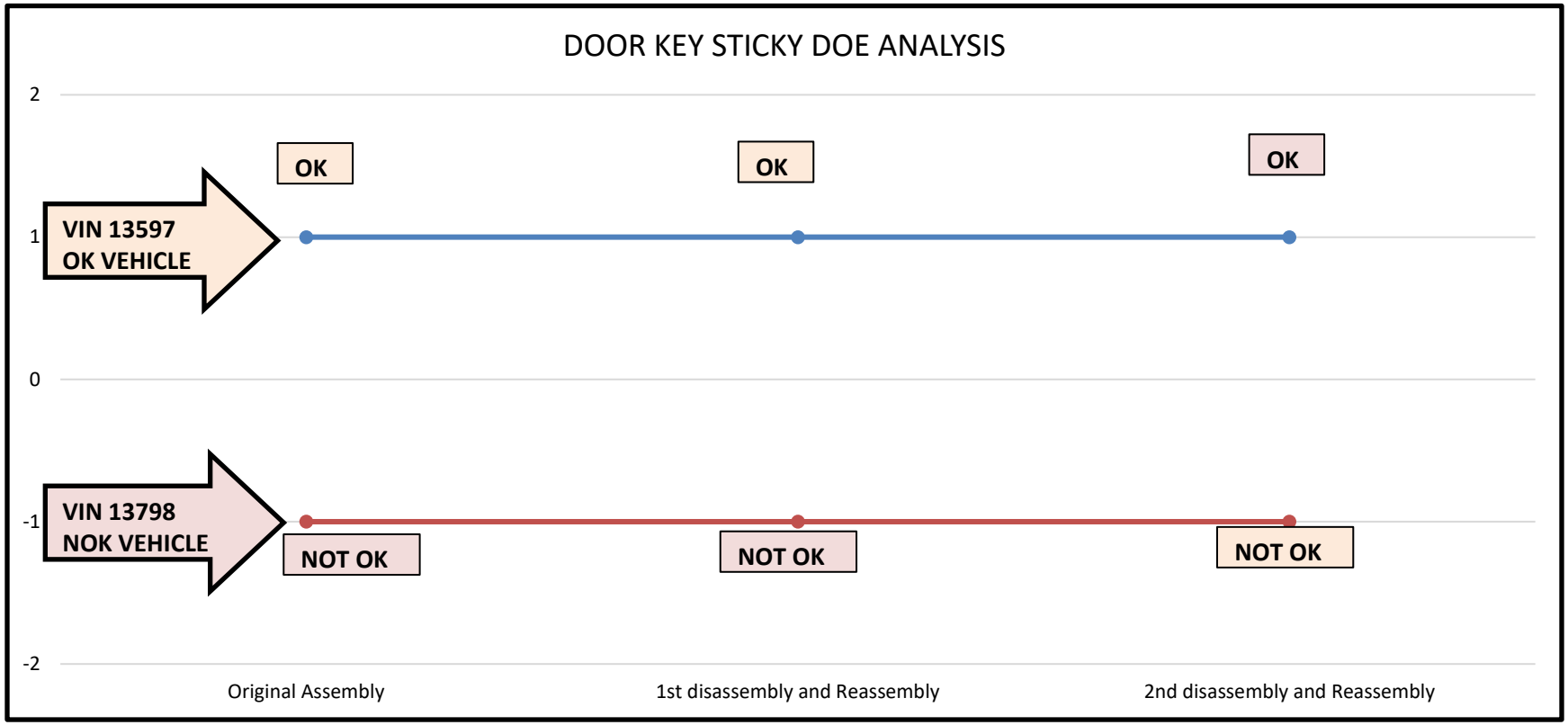
Response (Y) – Noise / No Noise

Rating	Criteria
OK	No door key sticky
Not OK	Door key sticky

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

A) Assembly Process Confirmation

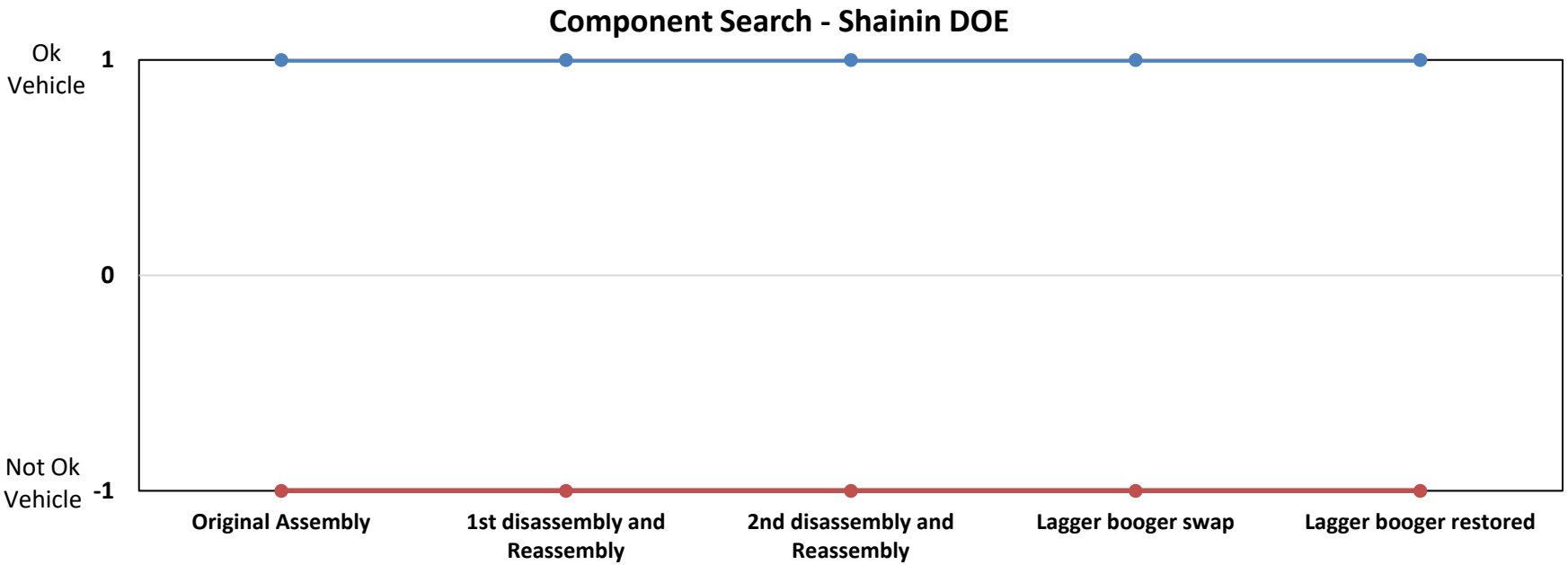


Inference :- Assembly Process is not a cause.

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

B) Lagger Booger

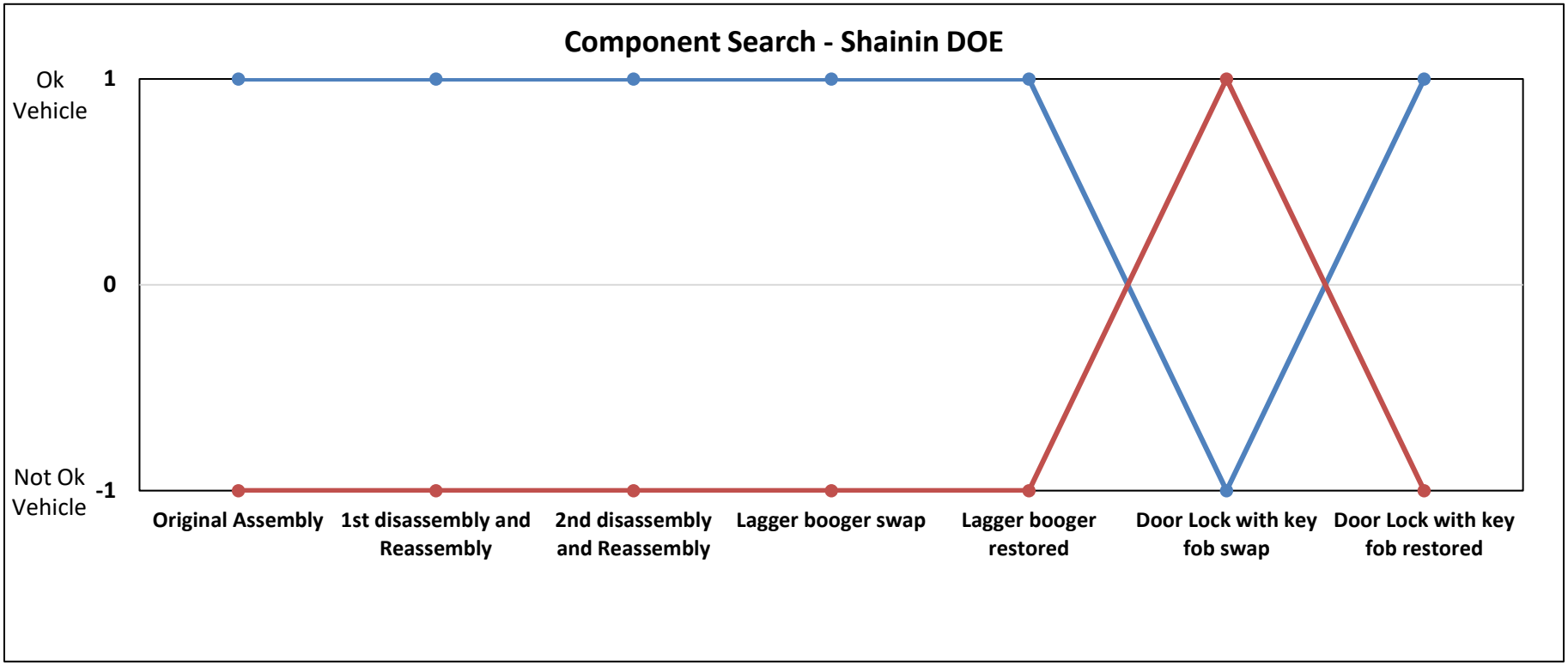


Inference :- Logger bugger part is not a cause.

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

C) Door lock with key fob



Inference :- Door lock with key fob is a cause.

5.Analysis

Observations on defective vehicle :-

NOT OK CONDITION

Reasons of **door key sticky** may be due to :-

A) Key fob – Ok & not Ok both vehicle key fob no major difference observed.

B) Door lock - Ok & Not Ok both vehicle door lock checked found circlip position disturbed.

5.1 List down SSV's or Xs

$$Y = f(x)$$

Response **SSV**

Y = Door key sticky
X1 = Door lock circlip position.

5.2 Identify the tool

Sr.No	SSV (X)	Drawing specifications
1	Door lock circlip position	Circlip should be at Center

The identified SSV's are Measurable on both Good and Bad components and are Input related Materials Also parameters can not be loss during measurement or not damage.

Hence DOE Tool

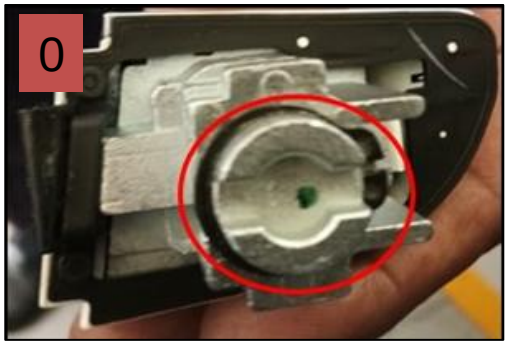
"Paired Comparison"

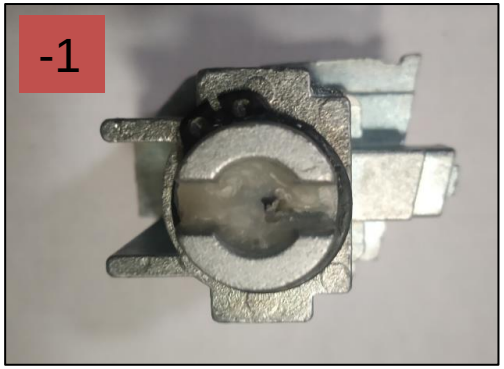
was selected.

5.5 Conclusions of Analysis

Vehicle	Circlip Position
B1	-1
B2	1
B3	1
B4	-1
B5	1
B6	1
B7	1
B8	1
W1	0
W2	0
W3	0
W4	0
W5	0
W6	0
W7	0
W8	0

- Data Collected for 8 Good & 8 Worst vehicle.
- On good vehicle there is no key sticky observed.
- On Worst vehicle key sticky observed.





Note :

- 0 considered as circlip position at centre.
- 1 & -1 considered circlip position on as LH/RH side of centre.

5.5 Conclusions of Analysis

Data arrange in ascending order :-

Vehicle	Circlip Position
B1	-1
B4	-1
B6	-1
W1	0
W2	0
W3	0
W4	0
W5	0
W6	0
W7	0
W8	0
B2	1
B3	1
B5	1
B7	1
B8	1

- Data Collected for 8 Good & 8 Worst vehicle.
- On good vehicle there is no key sticky observed.
- On Worst vehicle key sticky observed.

Inference:-

Top count – 3
Bottom count – 5
Total – 8

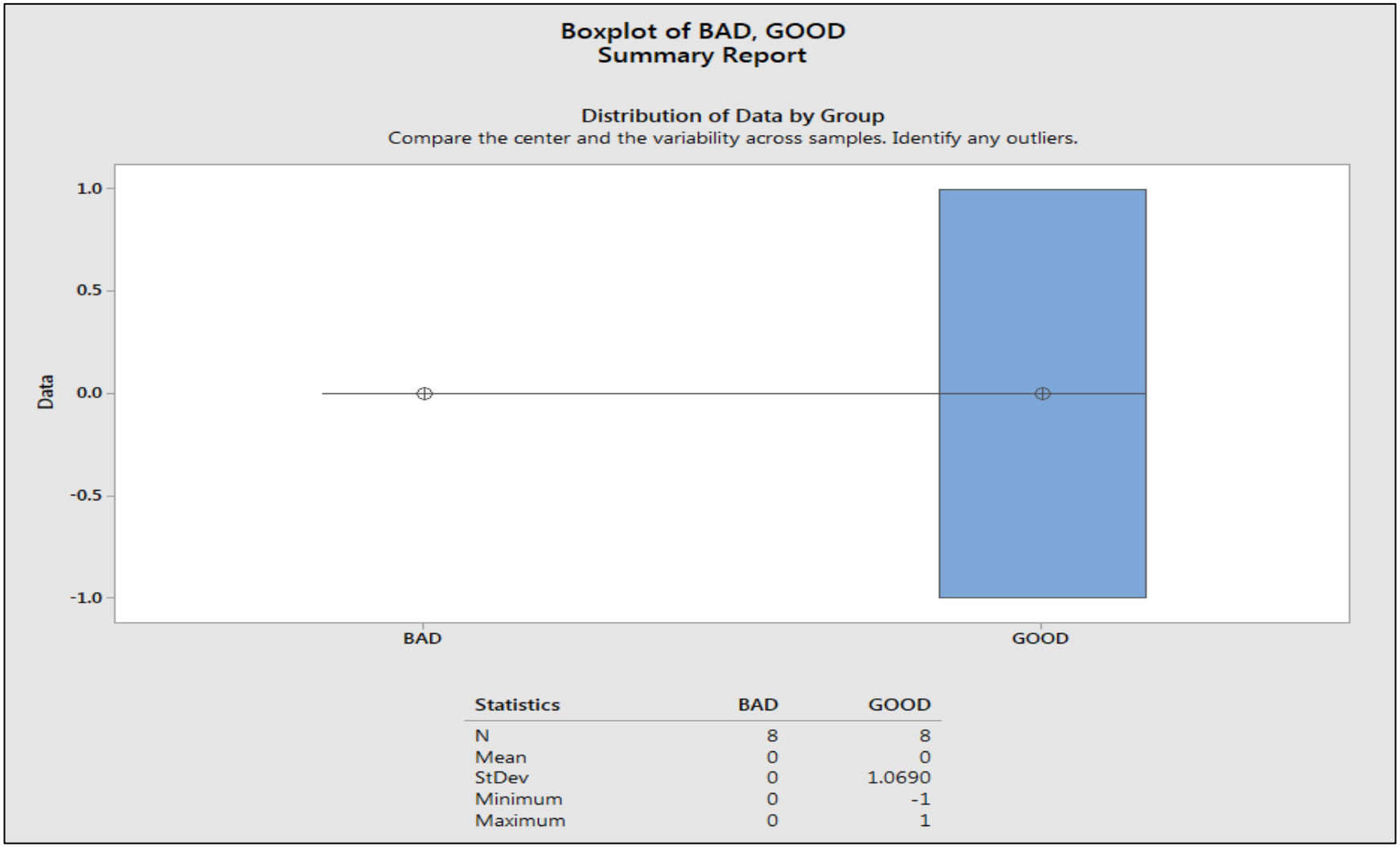
Rule 3 – Total count 8, so considering this parameter **is a confirm cause.**

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.5 Conclusions of Analysis

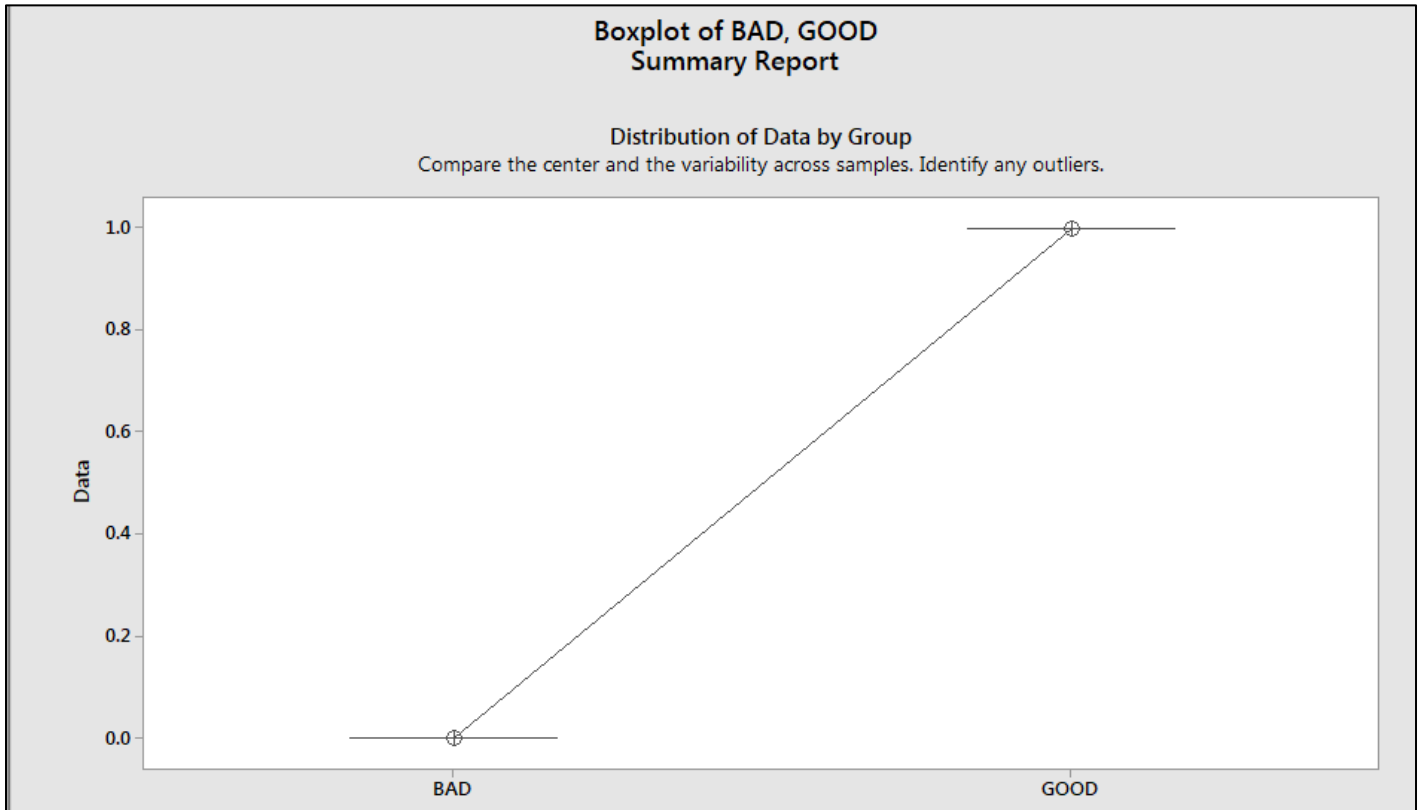
Box Plot for door lock circlip position :-



Statistics	BAD	GOOD
N	8	8
Mean	0	0
StDev	0	1.0690
Minimum	0	-1
Maximum	0	1

5.5 Conclusions of Analysis

Box Plot for door lock circlip position :-



Statistics	BAD	GOOD
N	8	8
Mean	0	1
StDev	0	0
Minimum	0	1
Maximum	0	1

5.5 Conclusions of Analysis

X	Probable Cause	Valid/ Invalid	Any changes in Spec.	X Controllable as per new spec Y/N	Drilling down required Y/N
X1	Door lock circlip position	Valid	NO	YES	No

Conclusion :-

Best Condition – When Door lock position not in center of slot.

Worst Condition – When Door lock position in center of slot.

5.6 Funneling to Next level causes

Door lock circlip fouls to lagger booger ribs

Why ?

Door lock circlip position found shifted

Why ?

No provision for circlip position.

Why ?

Circlip position not consider while design.

ACTION : Circlip position to be freeze.

DOE analysis

ROOT CAUSE

HOW`

Concern to be handover to PVT dept. for changes in current design.

COUNTER MEASURE

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

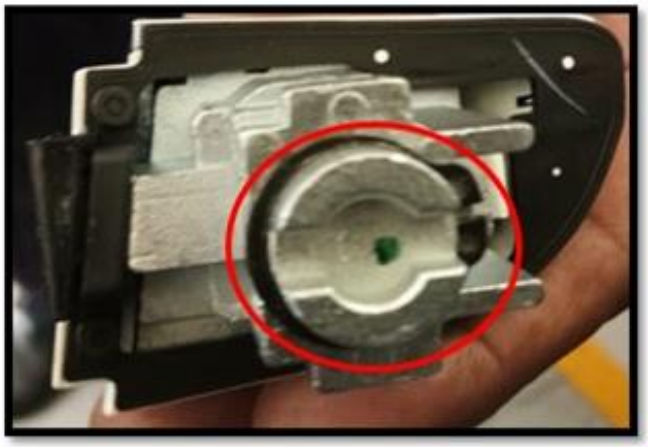
6.1 Generate possible solutions

X	Validated causes	Possible Solutions
X1	Circlip position variation	Circlip position to be freeze.

6.1 Generate possible solutions

Defect Simulation : Trial conducted with different circlip position

Key Cyllinder Black clip at center position



Key Cyllinder Black clip position shifted

Result

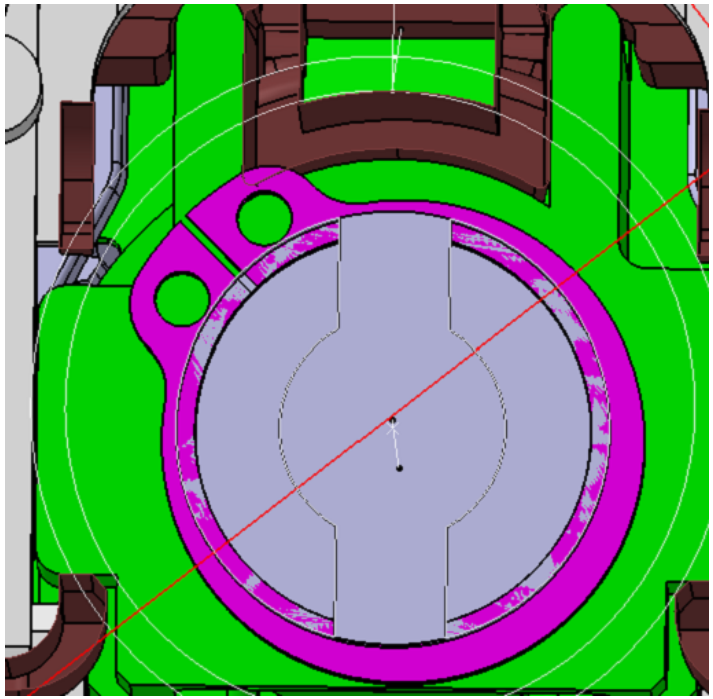
Sr. No.	Vehicle No.	Torque	Result
1	K2A13331	2Nm	Key Sticky
2	K2A13521	2Nm	Key Sticky
3	K2A13372	2Nm	Key Sticky
4	K2A13260	2Nm	Key Sticky
5	K2A13736	2Nm	Key Sticky
6	K2A13792	2Nm	Key Sticky

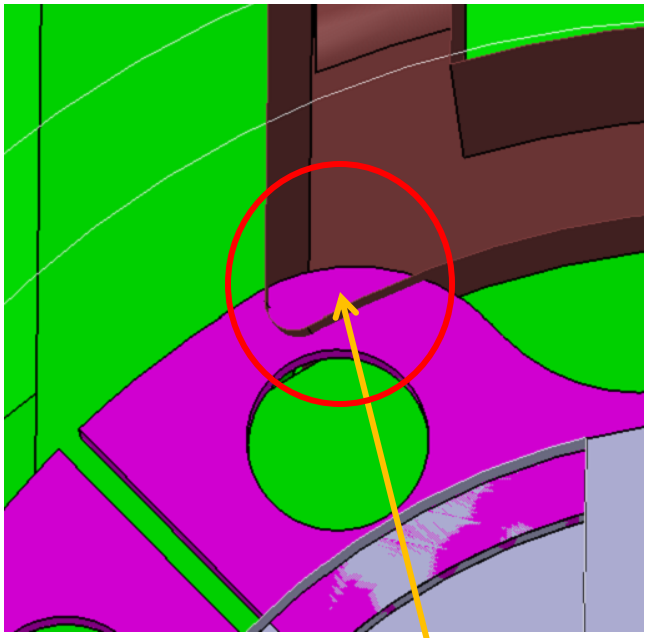
Result

Sr. No.	Vehicle No.	Torque	Result
1	K2A13331	2Nm	Key working OK
2	K2A13521	2Nm	Key working OK
3	K2A13372	2Nm	Key working OK
4	K2A13260	2Nm	Key working OK
5	K2A13736	2Nm	Key working OK
6	K2A13792	2Nm	Key working OK

Inference: If circlip position at centre & during operating it gets disturbed &creates defect of key sticky while retrieving.

6.1 Generate possible solutions



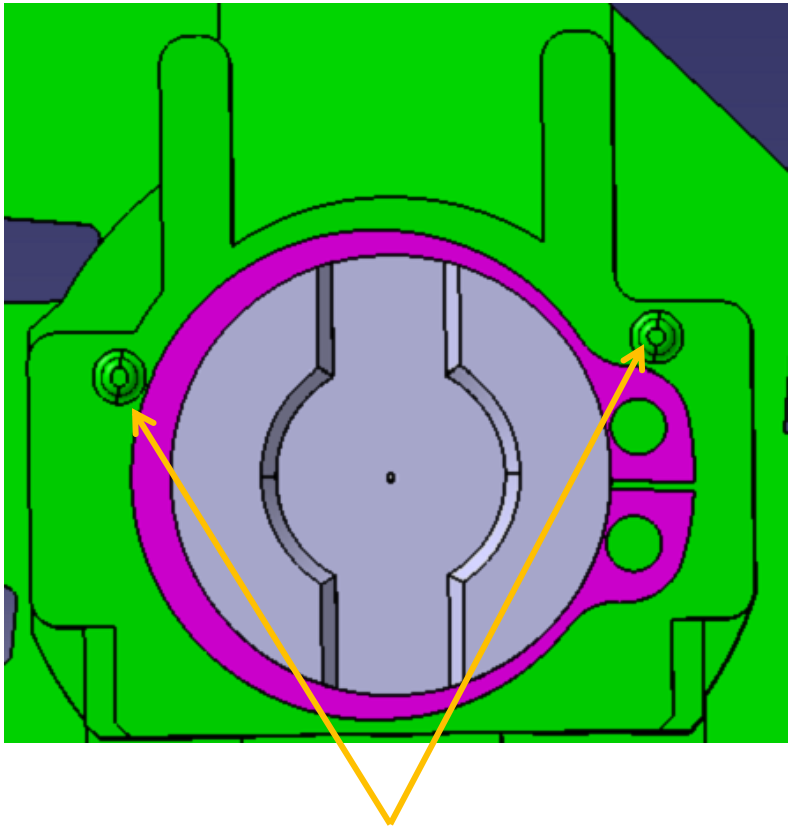


Bracket material will scratched by cir clip

Bracket material scratched by cir clip over wall (one edge)& cir clip will be jam

6.1 Generate possible solutions

Solution No.1 Stopper addition over stator for cir clip rotation



Stoppers for circlip

Minor modification required at stator tool

6.1 Generate possible solutions

Evaluation criteria

Effectiveness	
Solution's impact on root cause	**Rating**
Implementation of solution directly contributes to eliminate or control the root cause	10
Implementation of solution results in operating personnel capable of controlling the root cause	6 - 9
Implementation of solution results in creating awareness among operating personnel about the root cause- may result in reducing rejections	2 - 5
Implementation of solution does not result in controlling root cause	1

Time impact	
Solution implementation time	**Rating**
less than 15 days	10
15 to 20 days	7 - 9
20 to 30 days	4 - 6
greater than 30 days	2 - 3
Implementing solution will result in the increase in cycle time of manufacturing a piston	1

Cost / Benefit impact		
	Cost / Benefit category (both criteria to be satisfied)	**Rating**
(1)	One-time cost and also Cost incurred in implementing solution does not exceed department's sanctioned budgeted amount for the current year	10
(2)	Results in direct financial benefit	
(1)	One-time cost but Cost incurred in implementing solution exceeds department's sanctioned budgeted amount for the current year	7 - 9
(2)	Results in direct financial benefit	
(1)	Implementing solution will result in the increase of cycle time by not more than 25%	4 - 6
(2)	Results in indirect financial benefit	
(1)	Implementing solution will result in the increase of cycle time by more than 25%	2 - 3
(2)	Results in indirect financial benefit	
(1)	Implementing solution will result in direct loss of usable resource	1
(2)	Results in direct financial loss	

6.2 Evaluate and Select Solution

Solution Selection Matrix

	Effectiveness	Time Impact	Cost/Benefit Impact	<<<<Evaluation Criteria
	10	8	5	<<<<<<<<Importance
Solution	**Correlation of Solution to Criteria**			**Total**
Circlip position to be freeze.	10	8	10	214

6.3 Implement Solution

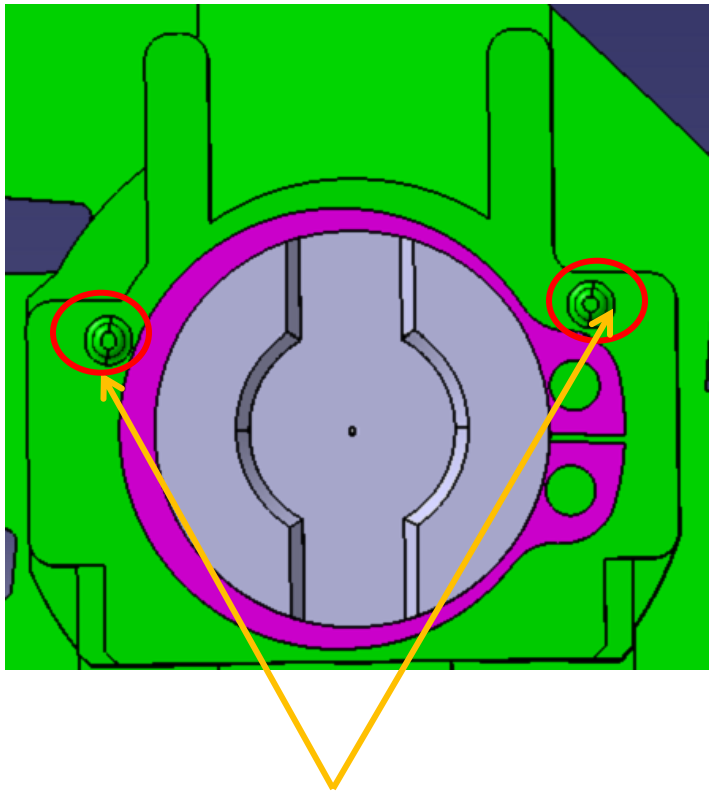
Solution Implementation Plan

X	Validated causes	Actions finalized	Resp.	Target Date	Status
X1	Circlip position variation.	Stopper addition over stator for cir clip rotation	Jitendra Baviskar	10.10.2019	Implemented.

CFT decided to implement solution i.e. circlip position to be freeze.by Stopper addition over stator for cir clip rotation.

6.3 Implement Solution

Circlip position freezed revised :-



Stoppers for circlip

7. Confirm Results and effects

1. Collect data
2. Capture all effects
3. Compare before and after
4. Measure Benefits

7.1 Collect data

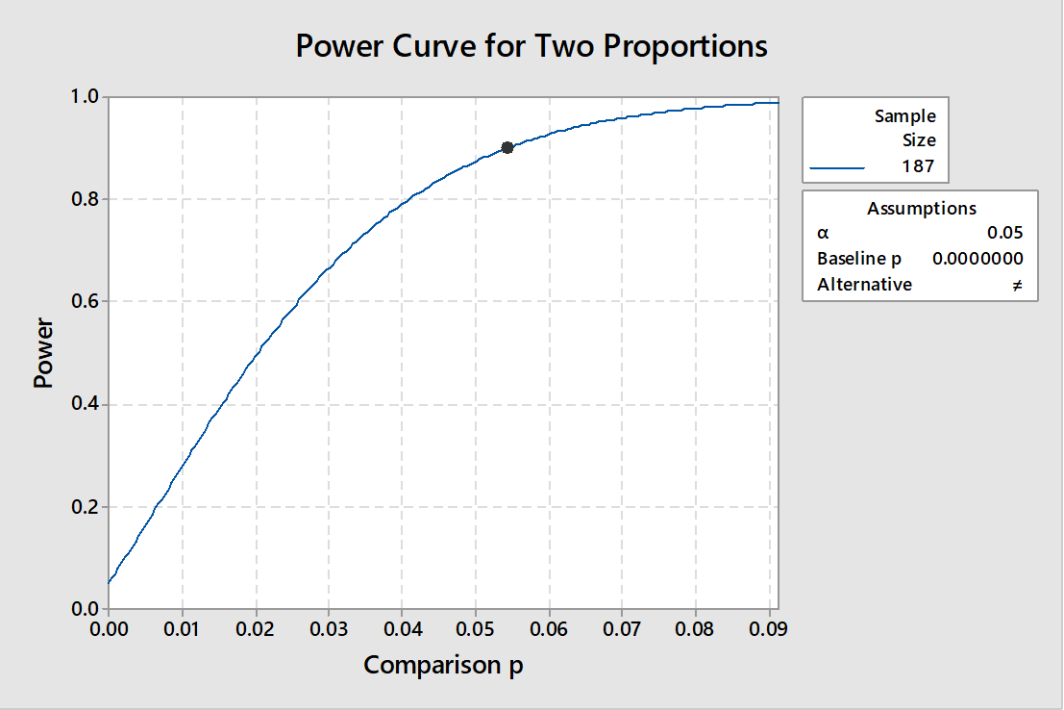
Data of after action – Sample size selection

```
Testing comparison p = baseline p (versus ≠)
Calculating power for baseline p = 0.000000009
α = 0.05


                Sample   Target
Comparison p     Size    Power   Actual Power
     0.05423      187      0.9       0.900894

The sample size is for each group.
```

Power Curve for Two Proportions



Inference: **It means we need to monitor 187 vehicles for concluding project is completed and required target is achieved.**

7.2 Capture all the effects

Effects on the selected problem:

- First time right launch achieved
- STR loss improved.
- Rework time saved.
- Rework manpower utilized for other reworks
- Vehicles passing to yard without delay.

Any other positive effects observed:

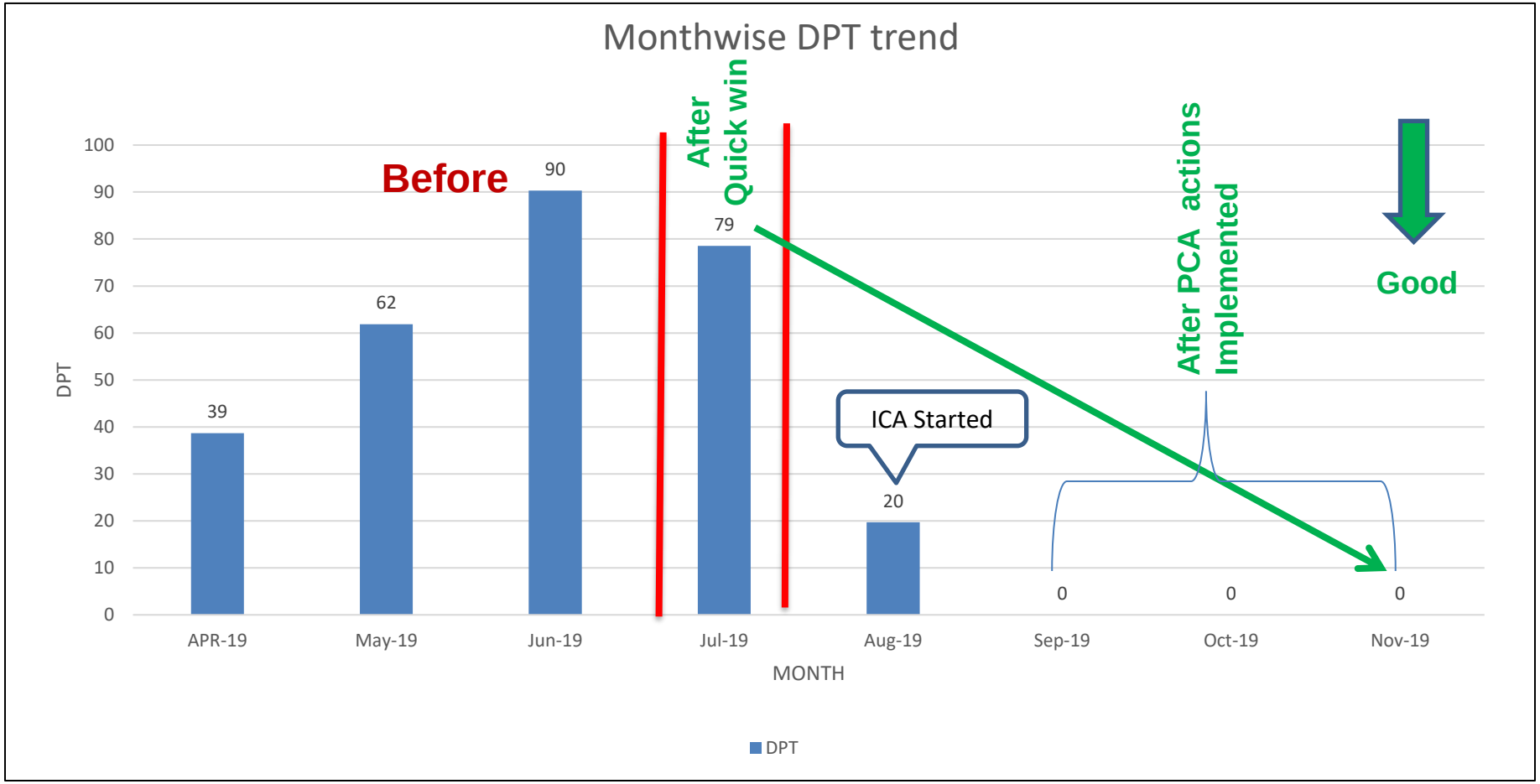
- CAI top repeating defect eliminated.
- Customer delight about Marazzo vehicle will increase.

Negative effects observed:

- NIL

7.3 Compare Before and After

Data of after action taken -



Inference: Defect rate of door key sticky noise is eliminated after permanent corrective actions.

7.3 Compare Before and After

Hypothesis testing for confirmation of results after PCA actions taken :-

Practical statement –

Defect rate of door key sticky (Y) was eliminated after taking corrective measures through PCA actions.

Statistical Problem –

Ho: Defect rate after corrective measure >= Defect rate before corrective measure.

Ha: Defect rate after corrective measure < Defect rate before corrective measure .

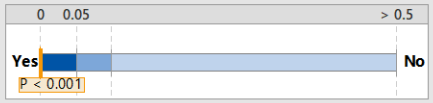
Analysis: 2P Test -

7.3 Compare Before and After

Analysis: 2P Test

2-Sample % Defective Test for Group 1 vs Group 2
Summary Report

Do the % defectives differ?



The % defective of Group 1 is significantly different from the % defective of Group 2 ($p < 0.05$).

Individual Samples

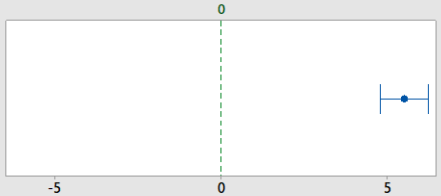
Statistics	Group 1	Group 2
Total number tested	4058	187
Number of defectives	223	0
% Defective	5.50	0.00
95% CI	(4.81, 6.24)	(0.00, 1.59)

Difference Between Samples

Statistics	*Difference
Difference	5.50
95% CI	(4.79, 6.20)

*Difference = Group 1 - Group 2

95% CI for the Difference
Is the entire interval above or below zero?



Comments

- Test: You can conclude that the % defective differs at the 0.05 level of significance.
- CI: Quantifies the uncertainty associated with estimating the difference from sample data. You can be 95% confident that the true difference is between 4.79% and 6.20%.

	Total vehicle produce	No of defects generated
Before action (Group 1)	4058	223
After action (Group 2)	187	0

Statistical Conclusion:
Reject the null hypothesis as P value observed is < 0.05

Practical Conclusion:
Defect rate of door key sticky (Y) was eliminated after taking corrective measures through PCA.

7.4 Measure Benefits

Tangible Benefits:

Sl No	Details	Cost in Rs
1	**Cost of poor quality per vehicle** a) Rework cost: Rs.10.48/- b) NIL c) NIL Total	Rs. 10.48/-
2	**Material saving or any other cost benefits per vehicle**	Rs. 10.0/-
3	**Productivity Benefits /any other benefits per vehicle**	**NA**
4	**No of vehicles produced per year (across sector) x Total Cost benefits per vehicle**	**Rs. 257806.45/-**

Intangible Benefits:

- Loss of STR avoided.
- Customer satisfaction (LSP , ANOVAC & Field)
- Quality indicators improvement.
- Through put time improved.
- Line flow become smooth due to no rework

7.4 Measure Benefits

Tangible Benefits:

No	Details	Cost in Rs
1	Cost of poor quality per vehicle a) Rework cost: Rs.292.47/-	Rs. 29.47/-
2	Material saving or any other cost	Rs. 0/-
3	Productivity Benefits /any other	0.22 Person saved
4	No of vehicles produced per year (across sector) x Total Cost benefits per vehicle	Rs.7,00,800/-

Intangible Benefits:

- Loss of FRC avoided.
- Quality indicators improvement.
- Through put time improved.
- Line flow become smooth due to no rework

8. Standardize

1. Update Documents
2. Train people to follow new standards
3. Audit standards

8.1 Update Document

Jump Seat Rattle Noise (Y1):

S N	Document	Responsibility	Status
1	ECN released.	Arvind Mistry	Closed
2	CAD Data updated.	Arvind Mistry	Closed

8.1 Update Document

Checfk arm noise (Y):- ECN copy

EN no. U321-EN-A003-000128_001

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000128
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 25-Jun-2019
Initiated Date 23-Jul-2019

Engineering Notice No : U321-EN-A003-000128

Released Date : 06-Aug-2019 14:06

Reference Change Req No : DCR043441

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : ADDITION OF STOPPER DIMPLE ON KEY BARREL HOLDER OF DRIVERDOOR LOCK

Reason : TO ADDRESS INTERNAL CONCERN OF KEY STICKY

Quantity : -(Regular)

Reason Code : Quality Improvement

Priority : Mandatory

Note : ADDITION OF STOPPER DIMPLE ON KEY BARREL HOLDER OF DRIVERDOOR LOCK

Affected Models : AU32BPES8T151A00SK, AU32CPES8T151A00SK, AU32EPES8T151A00SK, AU32EPES7T151A00SK, AU32FPES7T151A00SK, AU32BPES8T151A00XH, AU32CPES8T151A00XH, AU32EPES8T151A00XH, AU32EPES7T151A00XH, AU32FPES7T151A00XH, AU32BPES8T151A00CB, AU32CPES8T151A00CB, AU32EPES8T151A00CB, AU32EPES7T151A00CB, AU32FPES7T151A00CB, AU32BPES8T151A00WP, AU32CPES8T151A00WP, AU32EPES8T151A00WP, AU32EPES7T151A00WP, AU32FPES7T151A00WP, AU32BPES8T151A00GR, AU32CPES8T151A00GR, AU32EPES8T151A00GR, AU32EPES7T151A00GR, AU32FPES7T151A00GR, AU32BPES8T151A00PD, AU32CPES8T151A00PD, AU32EPES8T151A00PD, AU32EPES7T151A00PD, AU32FPES7T151A00PD, AU32BPES8T151A00ME, AU32CPES8T151A00ME, AU32EPES8T151A00ME, AU32EPES7T151A00ME, AU32FPES7T151A00ME

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : THIRUMALASETTI RAMESH (23077931)

Engineer : THIRUMALASETTI RAMESH (23077931)

Approved By : 207908-WANI CHANDRASHEKHAR

Authorized By : ELECTRICAL-AGG-GRP-HEAD_202176-KIRTY APURBO

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

Added/Deleted Qty : It's delta of existing released EN for the same parent part number

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

9. Summarize and Future Actions

1. Document Learning
2. Horizontal deployment
3. Future action

9. Summarize and Future Actions

9.1 Lessons Learnt :

1. CAD level study and design interface make the line chronic concern easy to get resolved
2. Assembly process observations in L0 analysis make critical role in defect analysis.
3. By considering all component from defect of door key sticky noise , exact root cause is established.
4. For any new product introduction study past trend history.
5. Involve supplier competency to get new way for analysis
6. DOE tools are very easy tool to identify the root cause & find out solution in a effective manner.
7. Take inputs from similar assembly process ongoing at some other location

9.2 Horizontal Deployment :

1. Actions taken to be horizontally deployed in other upcoming projects.

9.3 Future Actions :

1. NA.

Thanks !!!!!

- **Team members**
- **Mentors** : Mr.Rahul Narkar
- **MGB coordinator of commodity**: Mr. Ajay Namjoshi
- **MGB Coordinators**: Mr.Shrikant Mulay
- **MGB Tools Used** : Component Search ,Paired Comparison ,Full Factorial , Box Pot ,2P test etc.
- **HR for providing learning opportunity**.
- **MQS team**